



9/1/2026

N + - Network Fundamentals & Building Networks.

Network Topology Simulation.

TASK 1.

- ❖ Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

➤ Bus Topology:

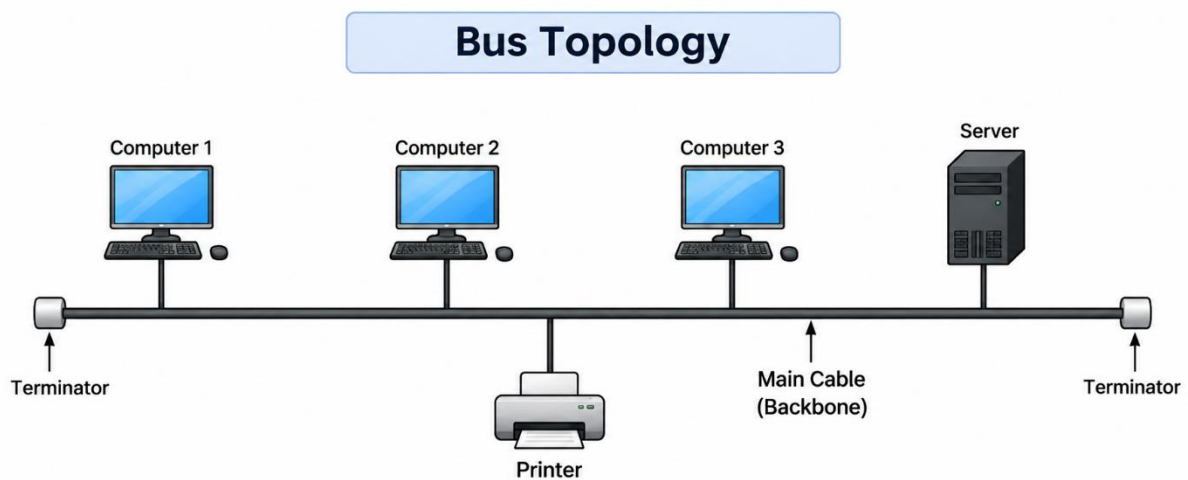


Figure : Bus Topology with Key Components.

Real-World Example:

A bus topology can be used in a **small office network**, where a few computers and a printer are connected through a single main cable.

TASK 2.

- ❖ In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

➤ Star Topology in Cisco Packet Tracer:

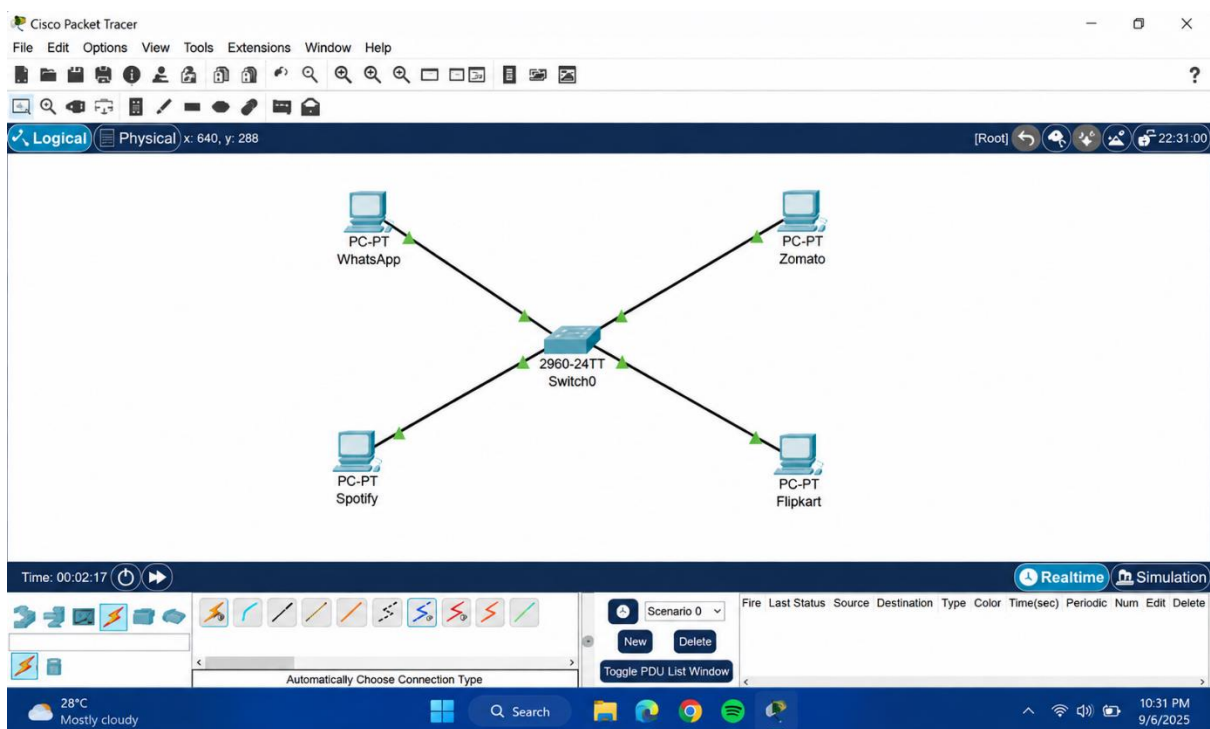


Figure : Star Topology with One Switch and Four PCs.

- ⌘ **Switch:** 2960-24TT Switch0
- ⌘ **PC 1:** WhatsApp
- ⌘ **PC 2:** Zomato
- ⌘ **PC 3:** Spotify
- ⌘ **PC 4:** Flipkart

All four PCs are directly connected to the **central switch**, forming a **star topology**.

TASK 3.

- ❖ Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup, Use arrows to show the direction of data flow.

➤ Ring Topology:

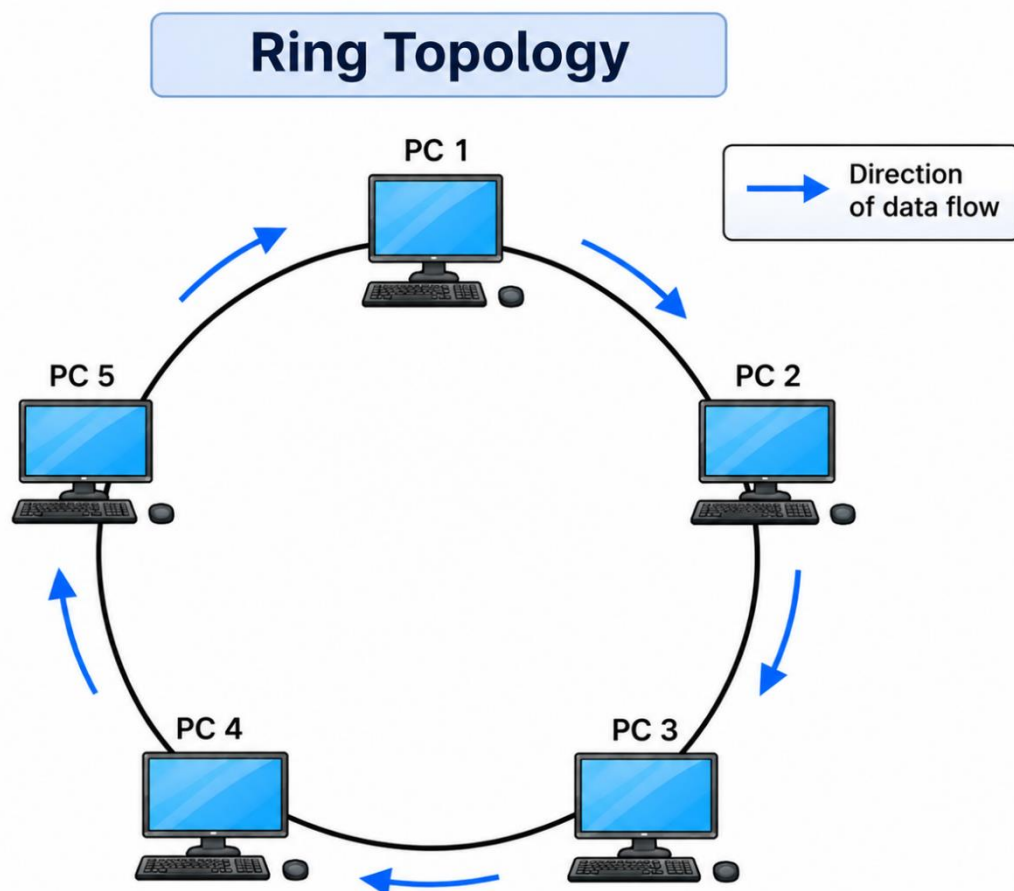


Figure : Ring Topology with Direction of Data Flow.

How Data Travels:

Data travels from one device to the next in a **single direction around the ring**, as shown by the arrows.

Each device receives the data and forwards it to the next device until it reaches the **destination device**.

TASK 4.

❖ **Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.**

➤ **Comparison of Mesh and Star Topologies:**

Topology	Advantages	Disadvantages
Mesh	1. Highly reliable because multiple paths are available. 2. Failure of one connection does not usually affect other devices.	1. Requires more cables and ports, making it expensive. 2. Difficult to install and maintain as the network grows.
Star	1. Easy to install, manage, and troubleshoot. 2. Failure of one device or cable does not affect the other devices.	1. Failure of the central switch can affect the entire network. 2. Requires more cable than some simpler topologies.